

from t_5 to $t_5 + \Delta t_5$ ($\Delta t_5 < \Delta t_2$), etc. Thus, a sequence of the seconds converging to the tangent line (drawn at point 5 of the graph $s(t)$) corresponds to a sequence of the average velocities converging to the slope α_5 of the tangent line, that is, to the value of the instantaneous velocity at the time moment t_5 .

Reader: It comes out that that the instantaneous velocity is the limit of a sequence of average velocities.

Author: Precisely. The instantaneous velocity is in fact the limit of a sequence of average velocities, provided that the time interval over which the averaging is made tends to zero converging to the moment of time t (viz, t_5 in Fig. 34).
Now let us formulate the definition in a more rigorous manner. What we want to define is the instantaneous velocity of a body at a moment of time t . Consider an arbitrary time interval from t to $t + \Delta t_1$. The distance covered by the body during this interval is Δs_1 . The average velocity of the body during this time interval is

$$V_{av}(t, \Delta t_1) = \frac{\Delta s_1}{\Delta t_1}$$

Next we select a shorter time interval Δt_2 from t to $t + \Delta t_2$ ($\Delta t_2 < \Delta t_1$) during which a distance Δs_2 is covered. Consequently, the average velocity over Δt_2 is

$$V_{av}(t, \Delta t_2) = \frac{\Delta s_2}{\Delta t_2}$$

We continue this process of selecting shorter and shorter time intervals starting at the moment of time t . As a result, we obtain a sequence of the average velocities

$$V_{av}(t, \Delta t_1), V_{av}(t, \Delta t_2), V_{av}(t, \Delta t_3), \dots$$

The limit of this sequence for $\Delta t \rightarrow 0$ is the instantaneous velocity at the moment of time t :

$$v(t) = \lim_{\Delta t \rightarrow 0} V_{av}(t, \Delta t) \quad (2)$$